RMO(INDIA) – 2013(1)

Max Mark – 102 Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.
- All Questions carry equal marks.
- Answer to each questions starts on a new page. Clearly indicate the question number.

1. Let ABC be an acute-angled triangle. The circle Γ with BC as diameter intersects AB and AC again at P and Q respectively. Determine $\angle BAC$ given that the orthocenter of triangle APQ lies on Γ.
2. Let $f(x) = x^3 + ax^2 + bx + c$ and $g(x) = x^3 + bx^2 + cx + a$, where a, b and c are integers with $c \neq 0$. Suppose that the following conditions hold:
   (a) $f(1) = 0$;
   (b) The roots of $g(x) = 0$ are the squares of the roots of $f(x) = 0$.
   Find the value of $a^{2013} + b^{2013} + c^{2013}$.
3. Find all primes p and q such that p divides $q^2 - 4$ and q divides $p^2 - 1$.
4. Find the number of 10 tuples $(a_1, a_2, ..., a_{10})$ of integers such that $|a_1| \leq 1$ and $a_1^2 + a_2^2 + ... + a_{10}^2 - a_1a_2 - a_2a_3 - ... - a_9a_{10} - a_{10}a_1 = 2$.
5. Let ABC be a triangle with $\angle A = 90^0$ and AB = AC. Let D and E be points on the segment BC such that $BD : DE : EC = 3 : 4 : 5$. Prove that $\angle DAE = 45^0$.
6. Suppose that m and n are integers such that both the quadratic equations $x^2 + mx - n = 0$ and $x^2 - mx + n = 0$ have integers roots. Prove that n is divisible by 6.

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